

Multiplying two large numbers together

If you are multiplying together two numbers that have more than one digit, things get a little trickier. Keep practising and you'll soon pick it up.



First concentrate on the unit digit at the bottom, and multiply it by each number on the top row in turn.

HTU
824
× 36

Ignore this 3 at first.
Multiply 6 by 4, then 2, then 8.

$$\begin{array}{r} 824 \\ \times 36 \\ \hline 4 \\ \hline 2 \end{array}$$

$6 \times 4 = 24$
carry the 2

$$\begin{array}{r} 824 \\ \times 36 \\ \hline 44 \\ \hline 12 \end{array}$$

$6 \times 2 = 12$
 $12 + 2 = 14$
carry the 1

$$\begin{array}{r} 824 \\ \times 36 \\ \hline 4944 \\ \hline 12 \end{array}$$

$6 \times 8 = 48$
 $48 + 1 = 49$

Now look at the tens digit at the bottom, and multiply it by the units, tens and hundreds digits in the top row. But first you need to add a zero, because you're multiplying by numbers in the tens column.

$$\begin{array}{r} 824 \\ \times 36 \\ \hline 4944 \\ 20 \\ \hline 1 \end{array}$$

$3 \times 4 = 12$
Add a zero
carry the 1

$$\begin{array}{r} 824 \\ \times 36 \\ \hline 4944 \\ 720 \\ \hline 1 \end{array}$$

$3 \times 2 = 6$
 $6 + 1 = 7$

$$\begin{array}{r} 824 \\ \times 36 \\ \hline 4944 \\ 24720 \\ \hline 1 \end{array}$$

$3 \times 8 = 24$

$$\begin{array}{r} 824 \\ \times 36 \\ \hline 4944 \\ + 24720 \\ \hline 29664 \end{array}$$

Finally, add together the two rows of numbers.

The answer is 29,664.

Now have a go

$$\begin{array}{r} 285 \\ \times 23 \\ \hline \end{array}$$

$$\begin{array}{r} 628 \\ \times 71 \\ \hline \end{array}$$

$$\begin{array}{r} 457 \\ \times 18 \\ \hline \end{array}$$

$$\begin{array}{r} 767 \\ \times 58 \\ \hline \end{array}$$

$$\begin{array}{r} 526 \\ \times 99 \\ \hline \end{array}$$

$$\begin{array}{r} 614 \\ \times 63 \\ \hline \end{array}$$

